

The inverse Hall–Littlewood $P \rightarrow m$ matrix is *not* in the
Wheeler–Zinn–Justin Hall–Littlewood papers
and the proposed DFK locator is edition-confused

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From: C. Vega (Clio) **To:** R. (grandpa-rick) **Date:** 2026-10-05
Re: your email of 2026-10-04 00:16, “*s=1 block valuation law; is the inverse HL $P \rightarrow m$ valuation known?*”
Source commit: clio-vega/work-in-progress@3f06cec
Grade: a *negative* literature finding, measured at first hand on two papers I hold, with controls.
 It is not a theorem.

1 Your question (a), answered: no

You asked whether the $t = 0$ case of your block valuation law — the $(1 - q)$ -adic valuation of $[h_\mu]Q'_\lambda(x; q)$, an entry of the inverse Hall–Littlewood $P \rightarrow m$ matrix, being $\ell(\lambda) - \kappa(\lambda, \mu)$ — appears in the Wheeler–Zinn–Justin Hall–Littlewood papers on my shelf.

Claim 1. *It is not in either of them, and the reason is structural rather than accidental: neither paper uses the monomial basis at all.*

I read both at first hand today. They are:

- Wheeler–Zinn–Justin, *Hall polynomials, inverse Kostka polynomials and puzzles*, [arXiv:1603.01815](#); journal edition *J. Combin. Theory Ser. A* **159** (2018) 107–163. (I give both editions deliberately; see §4.)
- Zinn–Justin, *Honeycombs for Hall polynomials*, [arXiv:1909.10720v2](#).

What they actually compute. WZJ study two one-parameter generalisations of $c_{\mu\nu}^\lambda$, and *both* live in bases other than m :

$$P_\mu P_\nu = \sum_\lambda f_{\mu\nu}^\lambda(t) P_\lambda \quad (\text{Hall polynomials; the } P \text{ basis}), \quad (1)$$

$$s_\mu P_\nu = \sum_\lambda \bar{K}_{\mu\nu}^\lambda(t) s_\lambda \quad (\text{their “generalized inverse Kostka”; the } s \text{ basis}). \quad (2)$$

Their $\bar{K}$ specialises at $\mu = 0$ to the expansion of P_ν in the **Schur** basis. That is the $P \rightarrow s$ matrix. The inverse of the $P \rightarrow m$ matrix is $m \rightarrow P$. These are different matrices, and the word “inverse Kostka” in their title refers to the former.

The measurement. In the full text of 1603.01815 (6662 extracted lines) the string `monomial symmetric` occurs **once**, at the eleventh line of the introduction, in the remark that Hall–Littlewood functions interpolate to monomial symmetric polynomials at $t = 1$. Every other occurrence of “monomial” in the paper means *a monomial in the lattice-model variables x_i* (“the coefficient of a single monomial in x ”, “a staircase monomial”), never a basis element m_μ . `monomial basis` and `m_{-}` occur zero times.

In 1909.10720 (3232 lines) the same holds: `monomial symmetric` occurs once, again as the $t = 1$ interpolation remark, and the only occurrence of `inverse Kostka` anywhere in the paper is **in the bibliography**, as the title of the WZJ reference. That paper proves a Pieri rule and associativity for a honeycomb formula for $f_{\mu\nu}^\lambda(t)$ — again purely the P basis.

Why I am willing to assert a null here. I have been wrong before by reporting a grep null that was an encoding artefact, so I ran the probe with controls. Positive controls in 1603.01815: Hall 55, Kostka 15, puzzle 61, Schur 33. In 1909.10720: Hall 43, honeycomb 61. A planted nonsense string returned 0. The search terms here are pure ASCII, so the en-dash class of false null does not apply. The nulls are real.

2 So your $t = 0$ edge is classical, not WZJ

Since $P_\lambda(x; 0) = s_\lambda$, at $t = 0$ your object degenerates to $[h_\mu]s_\lambda$, which is the Jacobi–Trudi / inverse-Kostka number — classical. The content of your law is therefore *not* at $t = 0$; it is the $(1 - q)$ -adic valuation statement at general q , where the $t = 0$ case is only the boundary value. I would frame the novelty claim that way round.

Pointers I have *not* verified at source. Offered as leads, not citations: Macdonald *Symmetric Functions and Hall Polynomials* §III.6 for the $P \rightarrow m$ transition matrix itself; Egecioglu–Remmel (1990) for the combinatorial inverse Kostka numbers via special rim-hook tabloids; and, closest to a *valuation* statement on Hall polynomials, Butler’s *Subgroup lattices and symmetric functions*, which does work p -adically. If $\ell(\lambda) - \kappa(\lambda, \mu)$ is in the literature I would expect it there or in the cocharge Kostka–Foulkes line (Lascoux–Schützenberger), not in the integrable-lattice line. Treat all four as unchecked.

3 Your question (b): the locator will not support the claim

You propose that Theorem C cite **DFK 1505.01657 Cor. 5.18** for the $t = 0$ edge, “whose $M_{k,1}$ is $E_k|_{t=0}$ literally”. That locator is edition-confused, and I resolved this by label-probe compile on 2026-10-04:

- In the **arXiv** edition of 1505.01657, the operator product $\chi_n = \prod M \cdots 1$ — the statement you want — is **Cor. 5.8**, p. 20.
- **Cor. 5.18**, p. 27, of that same edition is $\chi_n = \lim P_\lambda^{q,t}$, and contains **no M operator at all**.
- The “Corollary 18” cited in 1908.00806 is the **journal** numbering (*Transform. Groups* **23**(2), 2018), which equals arXiv Cor. 5.8. I believe that is where the number came from.

So: cite **Cor. 5.8** and name the edition, or cite “Cor. 18 (journal edition)”. As written, a referee checking the arXiv version finds a statement with no M in it. This is your own Risk 2 (“DFK15

normalization PENDING”); I consider it closed by the above, in your favour — the result you want *is* in DFK, under a different number.

4 What I have not done

I have not read §5 of 1603.01815 (their rank-two fermion–boson model for $\bar{K}$) in detail; only §1–§1.2, the definitions, and the closing normalisation. If you want the $\bar{K}_{\mu\nu}^\lambda(t)$ puzzle formula itself quoted precisely, ask and I will do that read properly. I have also not verified any of the four pointers in §2.